Data and Algorithm Analysis

Chapter 2 — Getting Started

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Outline

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Introduction

Sorting — Insertion Sort

Sorting — MergeSort

Traveling Salesperson Problem

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Administrative Details

- **Syllabus**
- **Textbook:** Introduction to Algorithms (Fourth Edition), by Cormen, Leiserson, Rivest, and Stein
- **Website:** https://thanghoang.github.io/teaching/f22/cs4104/
- **Canvas**
  - Website
  - Zoom links for office hours

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Overview

- Use frameworks for describing and analyzing algorithms.
- Examine two sorting algorithms: insertion sort and merge sort.
- Learn how to present an algorithm with *pseudocode*.
- Understand asymptotic notation for running-time analysis.
- Learn “divide and conquer” technique with merge sort.

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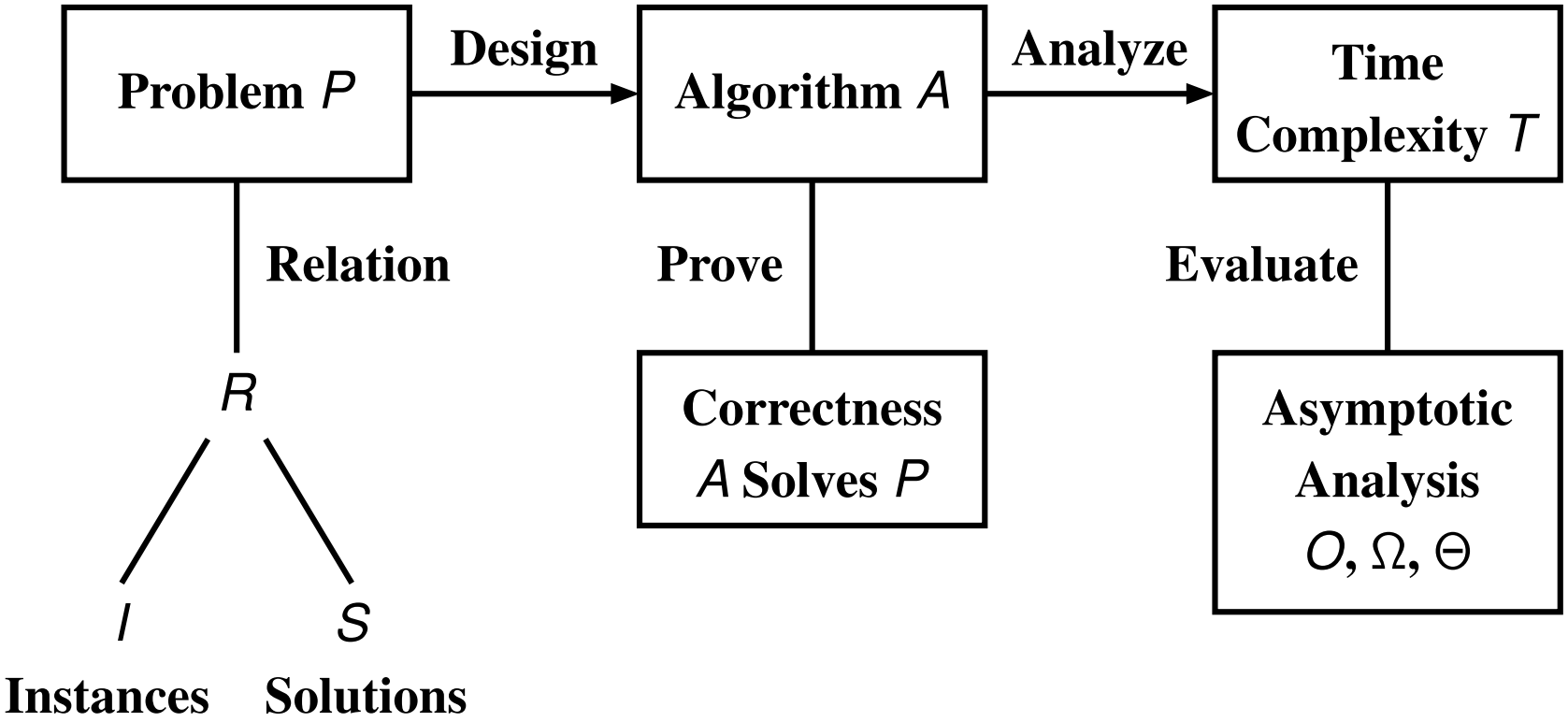


$$R \subseteq I \times S$$

Example — Sorting as a Formal Problem

SORTING

Input: Sequence $a_1, a_2, \ldots, a_n$ of integers.

Output: A permutation $b_1, b_2, \ldots, b_n$ of $a_1, a_2, \ldots, a_n$ such that

$$b_1 \leq b_2 \leq \cdots \leq b_{n-1} \leq b_n.$$

- For convenience of discussion, we often assume that the integers in the instance are distinct, though that is not strictly necessary.

Example — Sorting

An input of SORTING as an array $A[1:n]$:

$A[1]$	$A[2]$	$A[3]$	$A[4]$	$A[5]$	$A[6]$	$A[7]$	$A[8]$
57	23	31	100	4	18	44	92

The output of SORTING as an array $B[1:n]$:

$B[1]$	$B[2]$	$B[3]$	$B[4]$	$B[5]$	$B[6]$	$B[7]$	$B[8]$
4	18	23	31	44	57	92	100

We will now **design algorithms** to solve SORTING.

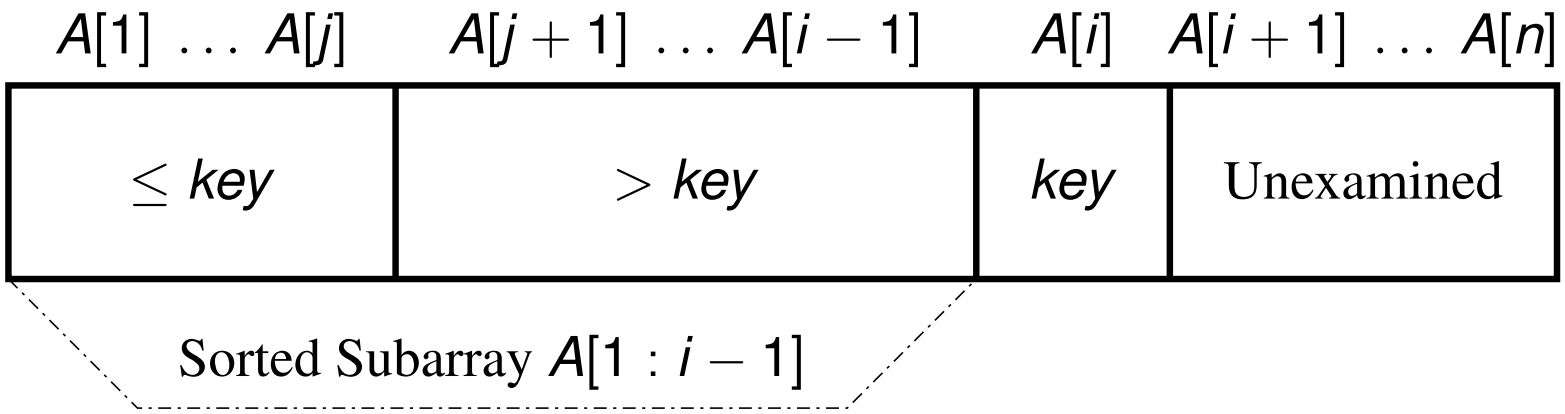
Insertion Sort — A Simple Sorting Algorithm

Main idea is to build up a sorted array in place as follows:

- Proceed through $A[1:n]$ iteratively from left to right.
- Always maintain a sorted subarray on the left of $A[i]$.
- At $A[i]$, find the right place to insert $A[i]$ into the sorted subarray to its left.
- In the process, move larger integers to the right.

Illustrating the Idea Behind Insertion Sort

$A[i] = key$ is the next integer to be inserted into the sorted subarray to its left.



We now express the INSERTION-SORT algorithm in **pseudocode.**

CLRS Pseudocode

- Emphasis on human readability and comprehension
- Indentation for logical structure
- Keywords: `if`/`else`/`elseif`, `while`, `for`, `repeat`/`until`, `return`
- Assignment: $=$
- Comment: //
- Lines numbered for reference purposes
- Flexible semantics

Pseudocode for INSERTION SORT

```
INSERTION-SORT(A, n)
    // A[1 : n] is an array of integers.
    // Returns a permutation of A in nondecreasing order.
    for i = 2 to n
        key = A[i]
        // Insert A[i] into the sorted subarray A[1 : i − 1].
        j = i − 1
        while j > 0 and A[j] > key
            A[j + 1] = A[j]
            j = j − 1
        A[j + 1] = key
    return A
```

Algorithm Design Paradigms

- Incremental — INSERTION SORT
- Divide and conquer — MERGESORT — recursive
- Dynamic programming — Chapter 14
- Greedy — Chapter 15
- Randomized algorithms — Chapters 5 and 7
- Exhaustive search — an approach, later, for the TRAVELING SALESPERSON PROBLEM

Insertion Sort — Correctness

Loop invariant: At the start of iteration i of the **for** loop (lines 3–10), the subarray $A[1 : i-1]$ is a sorted version of the original subarray $A[1 : i-1]$.

- **Initialization:** True before $i = 2$.
- **Maintenance:** True after loop body for i.
- **Termination:** Loop terminates with the array sorted when $i = n + 1$.

We conclude that the **for** loop actually sorts A.

INSERTION SORT — Verify Loop Invariant

```
INSERTION-SORT(A, n)
    // A[1 : n] is an array of integers.
    // Returns a permutation of A in nondecreasing order.
    for i = 2 to n          // A[1 : i − 1] is already sorted
        key = A[1]
        // Insert A[i] into the sorted subarray A[1 : i − 1].
        j = i − 1
        while j > 0 and A[j] > key
            A[j + 1] = A[j]
            j = j − 1
        A[j + 1] = key
    return A
```

Insertion Sort — Time Complexity Analysis

The final task is analyzing the time complexity of INSERTION-SORT.

This takes some discussion on analysis of algorithms, which is next.

Analysis of Algorithms

- Why analyze?
  - Why not just code the algorithm, run it, and time it?
  - Analysis tells you how long the code takes to run on different settings, inputs, or programming language.
- Random-Access Machine (RAM) model
- What to analyze in an algorithm?
  - Time complexity
    - Worst case
    - Average case
  - Space complexity

Worst Case Time Complexity

For algorithm A, *the worst case time complexity* $T_A(n)$ *of* A is the **maximum** number of computational steps taken by algorithm A on any instance of size n.

The parameter n is often clear from context; we will discuss it in more detail when we study the theory of NP-completeness.

Worst-case time gives a guaranteed **upper bound** on the running time for any input.

Average Case Time Complexity

Assume a probability distribution $f_n(I)$ on instances I of size n.

The time for instance I is $T(I)$.

For algorithm A, *the average case time complexity* $T(n)$ *of* A is the **average** number of computational steps taken by algorithm A on any instance of size n:

$$\begin{aligned} T(n) &= E[T(I) \mid \text{size of } I \text{ is } n] \\ &= \sum_{\text{size of } I \text{ is } n} f_n(I)T(I). \end{aligned}$$

Order of Growth

Abstraction to ease analysis and focus on the important features.

Look only at the leading term of the running time formula

- Drop lower-order terms
- Ignore the constant coefficient in the leading term

Example: $an^2 + bn + c$

Computing Worst Case $T(n)$

- Summation — for simple algorithms
- Recurrence — for divide and conquer algorithms

INSERTION SORT — Just the **for** Loop

cost	*times*		
c_3	n	3	**for** $i = 2$ **to** n
c_4	$n-1$	4	$key = A[1]$
c_6	$n-1$	6	$j = i - 1$
c_7	$\sum_{i=2}^{n} t_i$	7	**while** $j > 0$ and $A[j] > key$
c_8	$\sum_{i=2}^{n} t_i - 1$	8	$A[j+1] = A[j]$
c_9	$\sum_{i=2}^{n} t_i - 1$	9	$j = j - 1$
c_{10}	$n-1$	10	$A[j+1] = key$

To get $T(n)$, need to multiply *cost* $\times$ *times* and sum it all up.

Insertion Sort — Computing $T(n)$

$$
\begin{aligned}
T(n) &= c_3 n + c_4(n-1) + c_6(n-1) + c_7\left(\frac{n(n+1)}{2} - 1\right) \\
&\quad + c_8\left(\frac{n(n-1)}{2}\right) + c_9\left(\frac{n(n-1)}{2}\right) + c_{10}(n-1) \\
&= \left(\frac{c_7}{2} + \frac{c_8}{2} + \frac{c_9}{2}\right) n^2 \\
&\quad + \left(c_3 + c_4 + c_6 + \frac{c_7}{2} - \frac{c_8}{2} - \frac{c_9}{2} + c_{10}\right) n \\
&\quad - (c_4 + c_6 + c_7 + c_{10})
\end{aligned}
$$

So, $T(n)$ is a **quadratic polynomial.**

Asymptotics of $T(n)$

- Asymptotics are expressed using **asymptotic notation,** including Θ.
- For INSERTION SORT, we have $T(n) = \Theta(n^2)$; more details on asymptotic notation in Chapter 3.
- That completes the analysis of INSERTION SORT.

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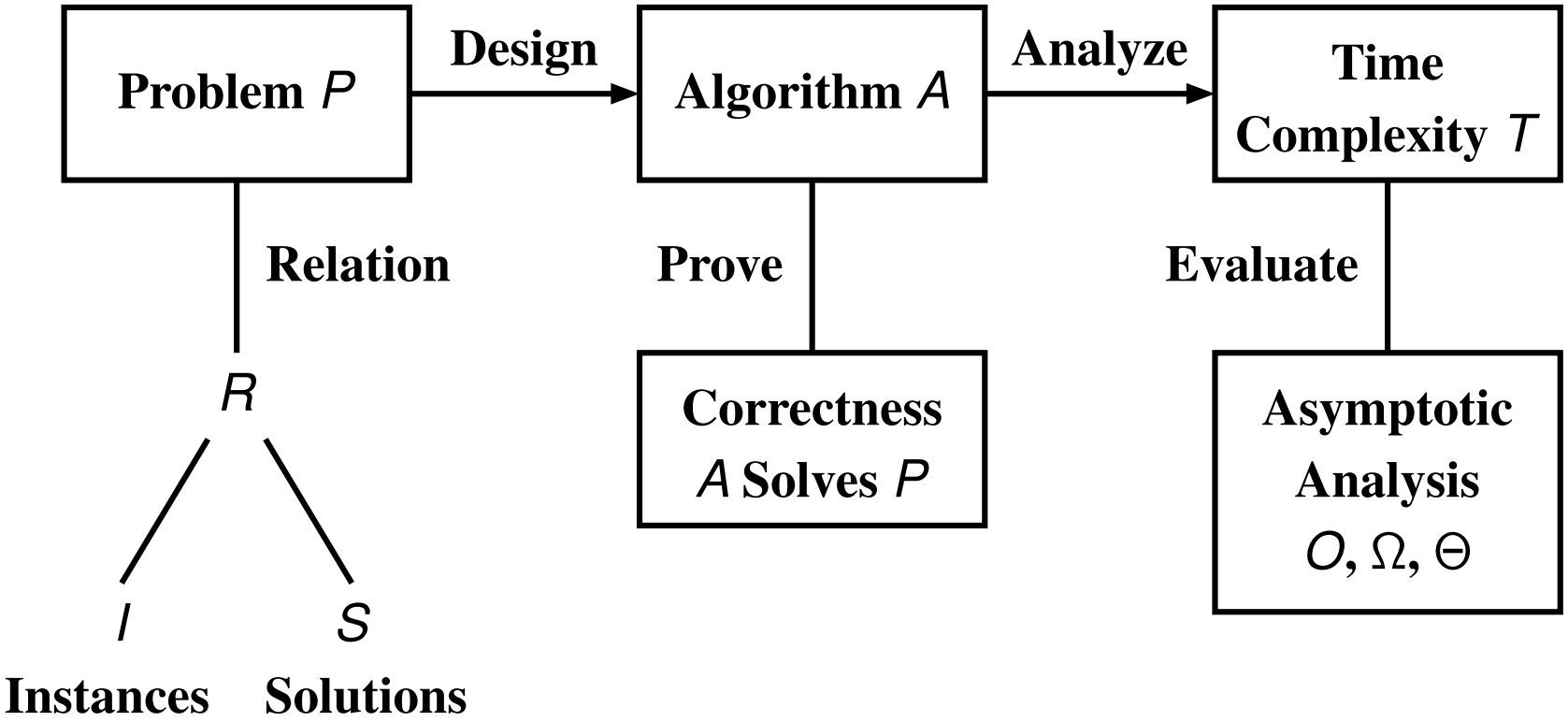

$$R \subseteq I \times S$$

MergeSort — An Alternate Sorting Algorithm

- MergeSort: divide and conquer algorithm
- Recursive algorithm MERGE-SORT(A, p, r)
- Represents instance as an array $A[1 : n]$
- Initial call MERGE-SORT$(A, 1, n)$
- **Divide and Conquer** — Splits array in two and recursively sorts each subarray
- Uses a MERGE algorithm to merge two sorted subarrays into one sorted array
- Pseudocode follows

MERGE-SORT Algorithm

```
MERGE-SORT(A, p, r)
    // A[p : r] = a_p, a_{p+1}, ..., a_r is an array of integers.
    // Returns an in-place permutation of A[p : r]
    //        in non-decreasing order.
    if p < r
        q = ⌊(p + r)/2⌋
        MERGE-SORT(A, p, q)
        MERGE-SORT(A, q + 1, r)
        MERGE(A, p, q, r)
```

MERGE Algorithm

```
MERGE(A, p, q, r)
    // A[p : r] = a_p, a_{p+1}, ..., a_r is an array of integers
        // with sorted subarrays A[p : q] and A[q + 1 : r].
    // Returns an in-place permutation of A[p : r]
    //        in non-decreasing order.
    n_1 = q − p + 1
    n_2 = r − q
    let L[1 : n_1 + 1] and R[1 : n_2 + 1] be new arrays
    for i = 1 to n_1
        L[i] = A[p + i − 1]
    for j = 1 to n_2
        R[j] = A[q + j]
```

MERGE Algorithm (Continued)

```
L[n_1 + 1] = ∞
R[n_2 + 1] = ∞
i = 1
j = 1
for k = p to r
    if L[i] ≤ R[j]
        A[k] = L[i]
        i = i + 1
    else A[k] = R[j]
            j = j + 1
```

Time Complexity Analysis

- The MERGE algorithm (conquer step) takes linear time.
- The divide step takes at most linear time.
- So, the nonrecursion time at any call to MERGE-SORT is at most $c_1 n$, for some positive constant c_1.
- For a call to MERGE-SORT that does not result in recursion, we say the time complexity is some other positive constant c_2.

Recurrences for Time Complexity

- MergeSort recurrence:

$$T(n) = \begin{cases} 2T(n/2) + c_1 n & n > 1 \\ c_2 & n = 1 \end{cases}$$

- Solving recurrence by recursion tree yields

$$T(n) = c_1 n \lg n + c_2 n.$$

- Details on recursion trees in Chapter 4.
- **Asymptotics —** $T(n) = \Theta(n \lg n)$

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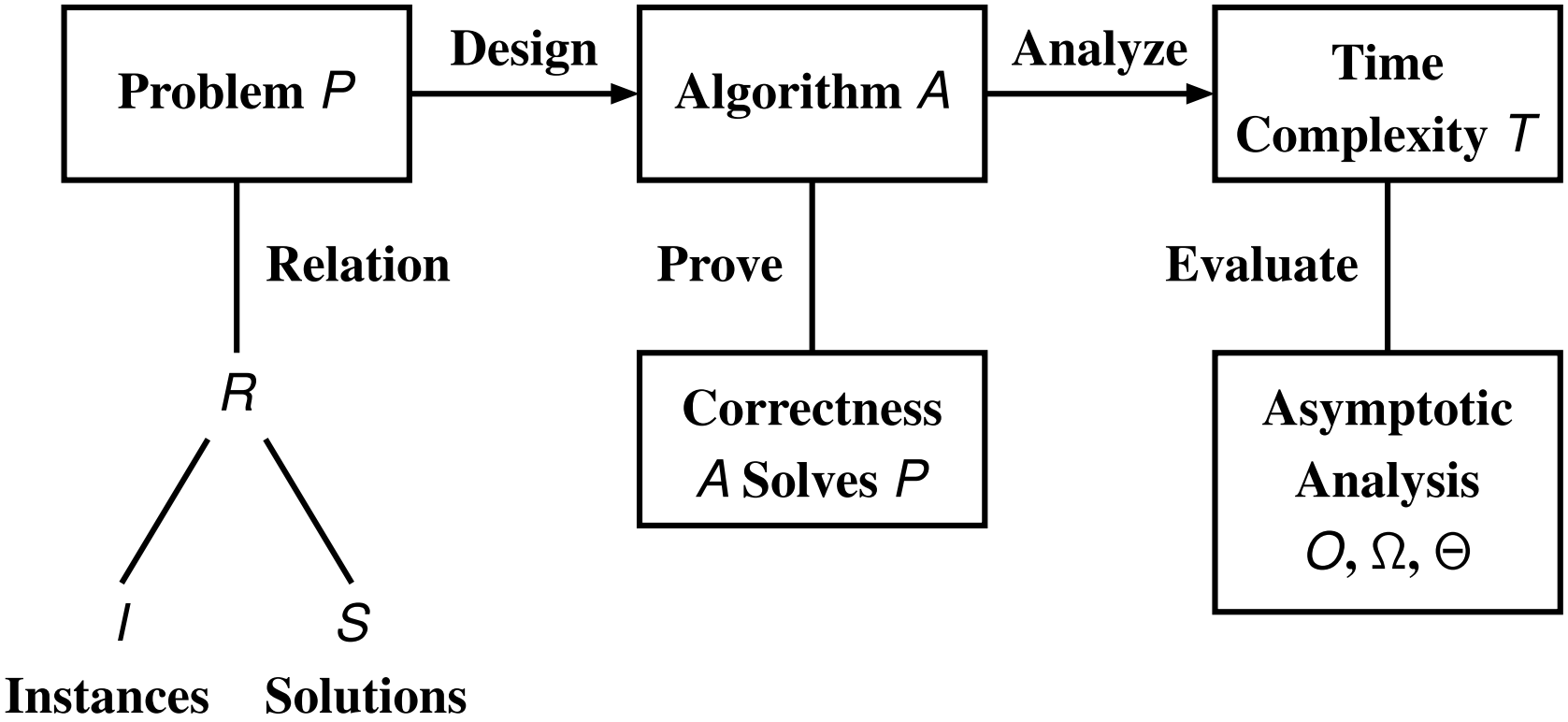


$$R \subseteq I \times S$$

Bonus Example — Traveling Salesperson Problem

TRAVELING SALESPERSON PROBLEM (TSP)

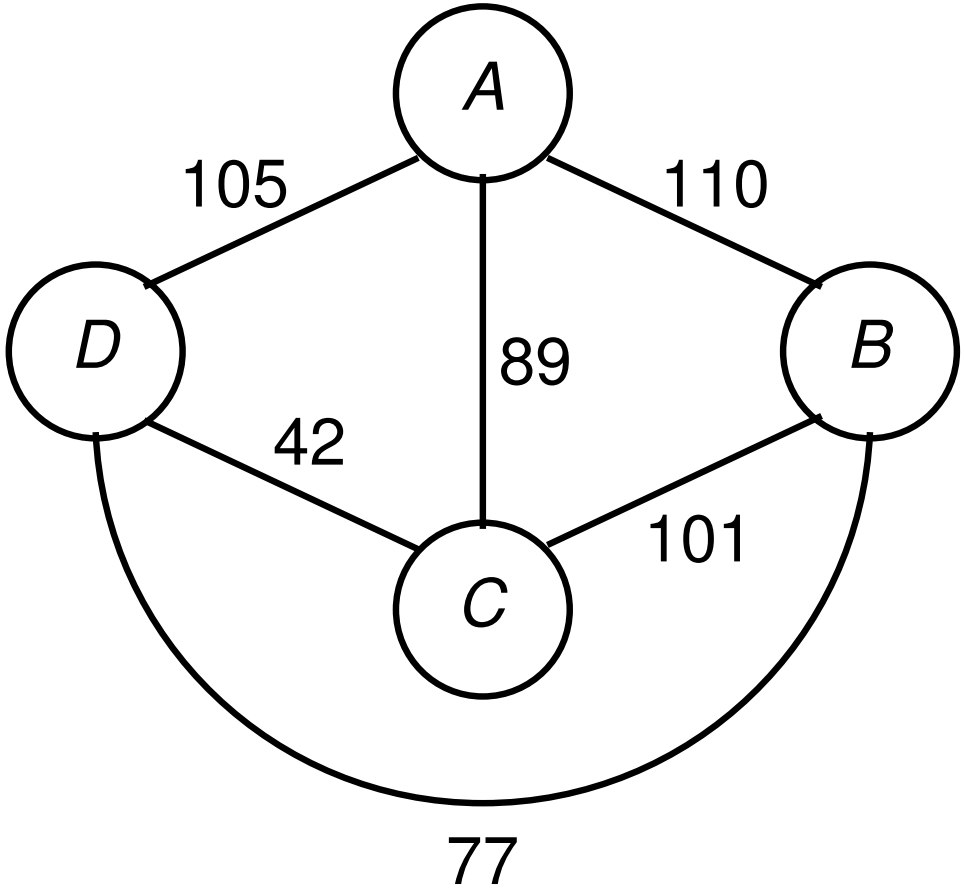
Input: Complete undirected graph $G = (V, E)$; weight function $w : E \to \mathbb{Z}$.

Output: A permutation $v_1, v_2, \ldots, v_n$ of V such that

$$w(v_n, v_1) + \sum_{i=1}^{n-1} w(v_i, v_{i+1})$$

is minimized.

Example — TSP Instance



Algorithm Design for TSP

Exhaustive search — an approach to the TRAVELING SALESPERSON PROBLEM

- Given $G = (V, E)$ and $w : E \to \mathbb{Z}$
- Generate every permutation of V
- Compute weight of each
- Return permutation of minimum weight

Pseudocode for TSP-EXHAUSTIVE

```
TSP-EXHAUSTIVE(G, w)
    // G = (V, E) is a complete undirected graph.
    // w : E → ℤ is an edge weight function.
    // Returns a permutation of V of minimum total weight.
    s* = ∞          // minimum weight so far
    π* = NIL        // permutation of weight s*
    for π = v_1, v_2, ..., v_n a permutation of V
        s = w(v_n, v_1) + Σ_{i=1}^{n-1} w(v_i, v_{i+1})
        if s < s*
            s* = s
            π* = π
    return π*
```

Time Complexity

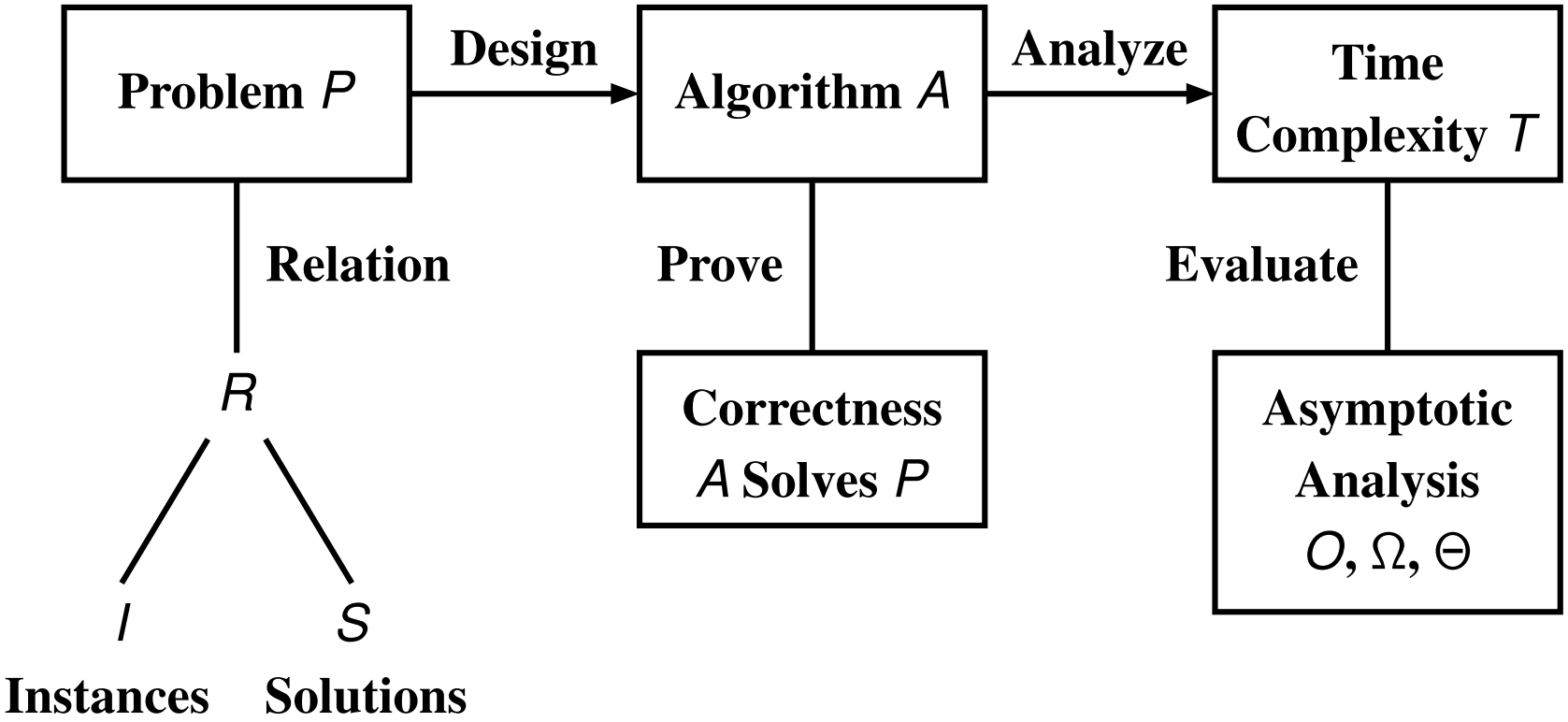
- **for** loop executed $n!$ times
- Line 7: summation takes $c_1 n$ operations for a positive constant c_1
- Generate next permutation in constant time c_2; see papers by Heap and Sedgewick under Resources
- Total time:

$$T(n) = n!(c_1 n + c_2) + c_3$$

- Asymptotics:

$$T(n) = O(n \cdot n!)$$
$$T(n) = \Omega(n \cdot n!)$$
$$T(n) = \Theta(n \cdot n!)$$

Algorithms Solve Problems



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